

IUS
INSTITUT
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DES SCIENCES

Institut Universitaire des Sciences

Prénom : Jonathan K.

Nom : SYLVAINCE

Faculté : FST

Niveau : L3

TD4 Réseaux

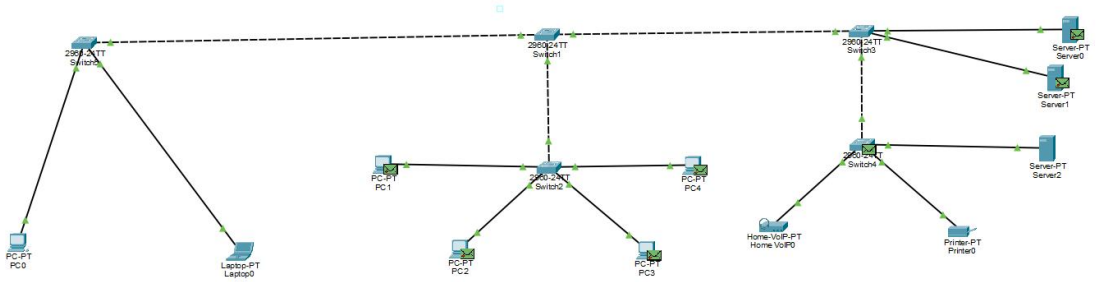
Date : 13 novembre 2026

**L’objectif de ce TD est de :**

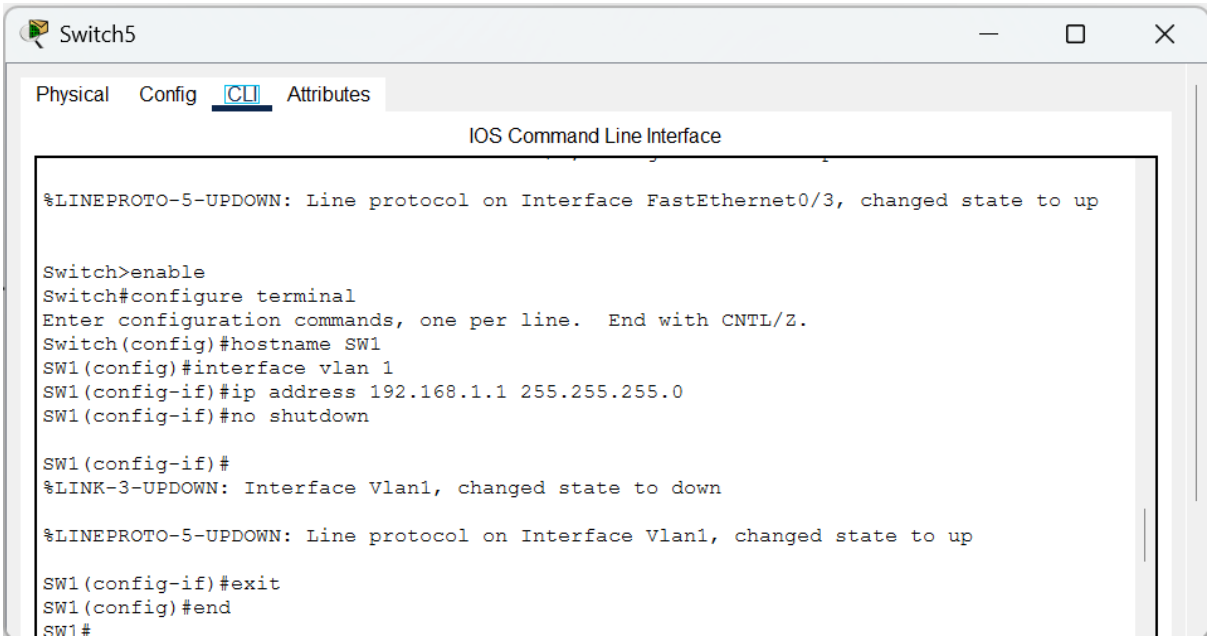
Ce TD avait pour but de découvrir la configuration de base d'un switch et d'un routeur avec Cisco Packet Tracer, d'attribuer des adresses IPv4 et IPv6 aux équipements (switchs, routeur, PC), de tester la connectivité entre les hôtes avec la commande ping, et d'utiliser le mode Simulation pour observer et analyser le trafic réseau (paquets ICMP, ARP).

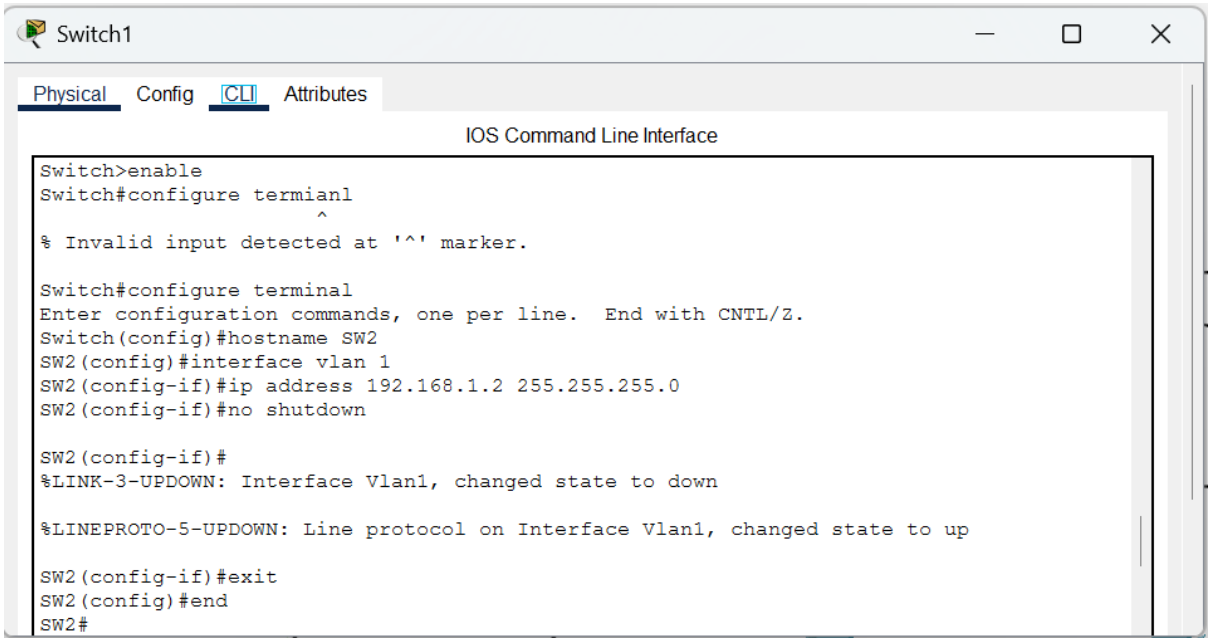
**Travaux Dirigés**

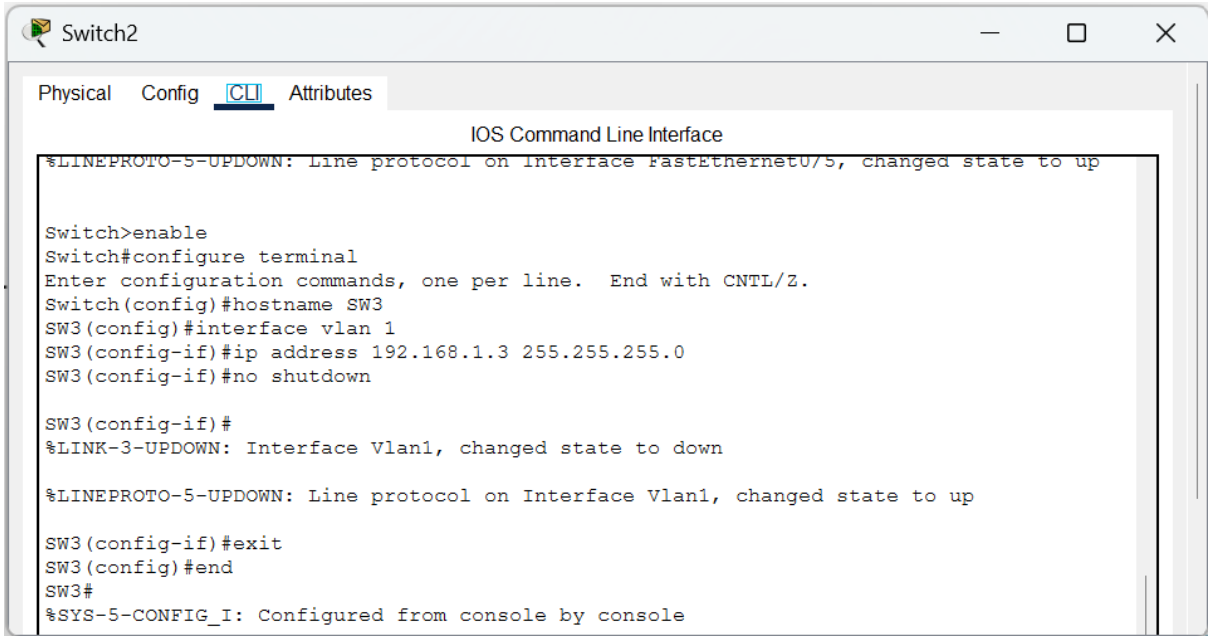
1- Reproduisez cette topologie en configurant les switchs, puis en attribuant les adresses IP aux dispositifs. Utilisez soit IPv4, soit IPv6, et testez la connectivité des deux VLAN à l’aide de la commande ping et du mode de simulation.

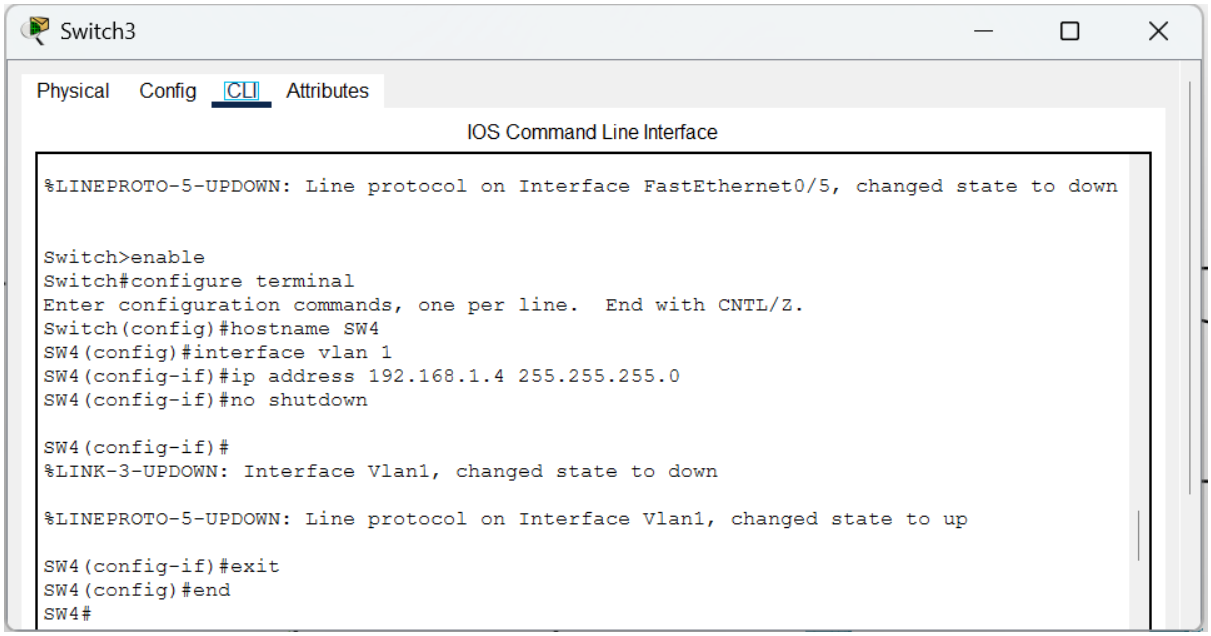


Configuration des switchs IPv4 :

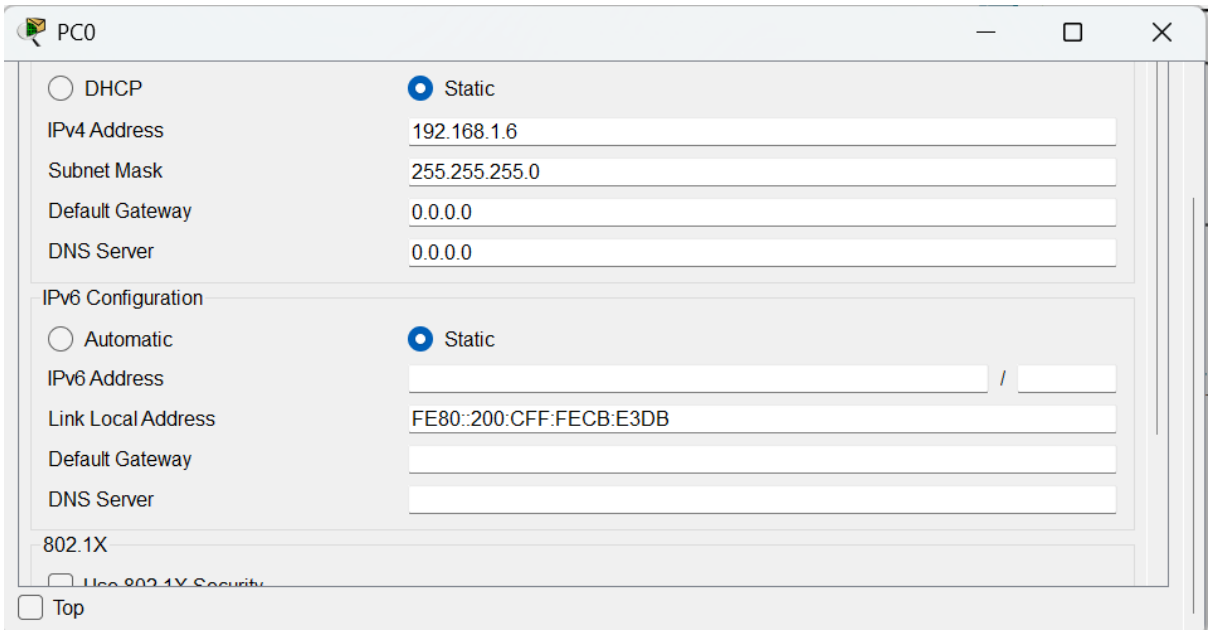


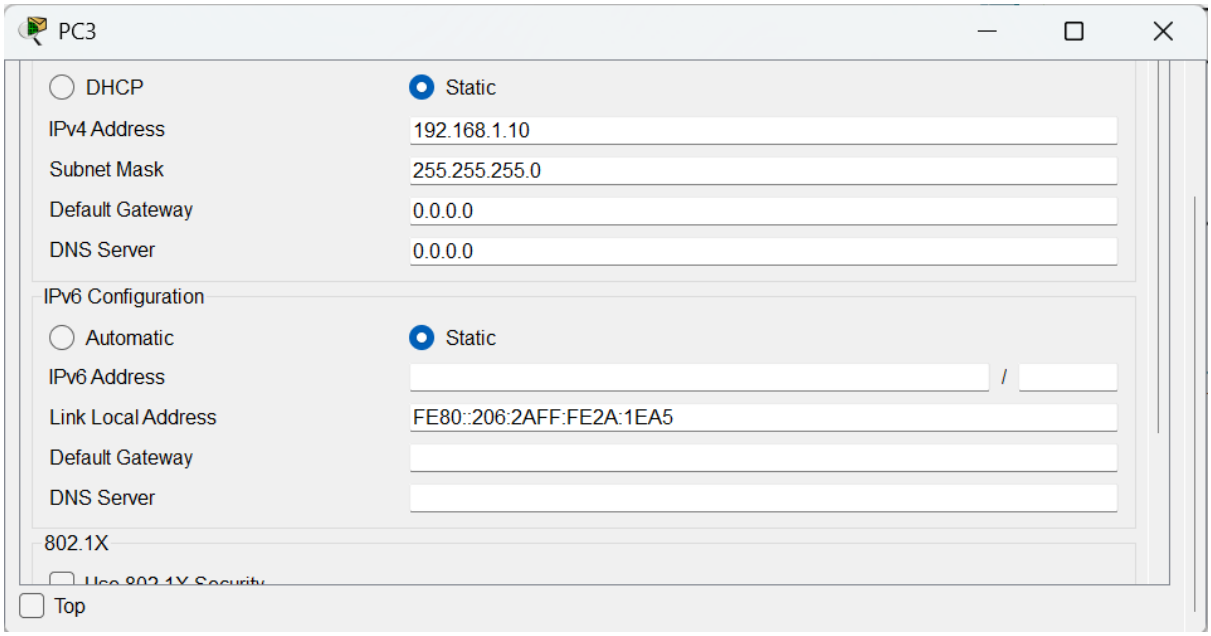

Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>enable
Switch#configure termianl
^
% Invalid input detected at '^' marker.
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW2
SW2(config)#interface vlan 1
SW2(config-if)#ip address 192.168.1.2 255.255.255.0
SW2(config-if)#no shutdown
SW2(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
SW2(config-if)#exit
SW2(config)#end
SW2#

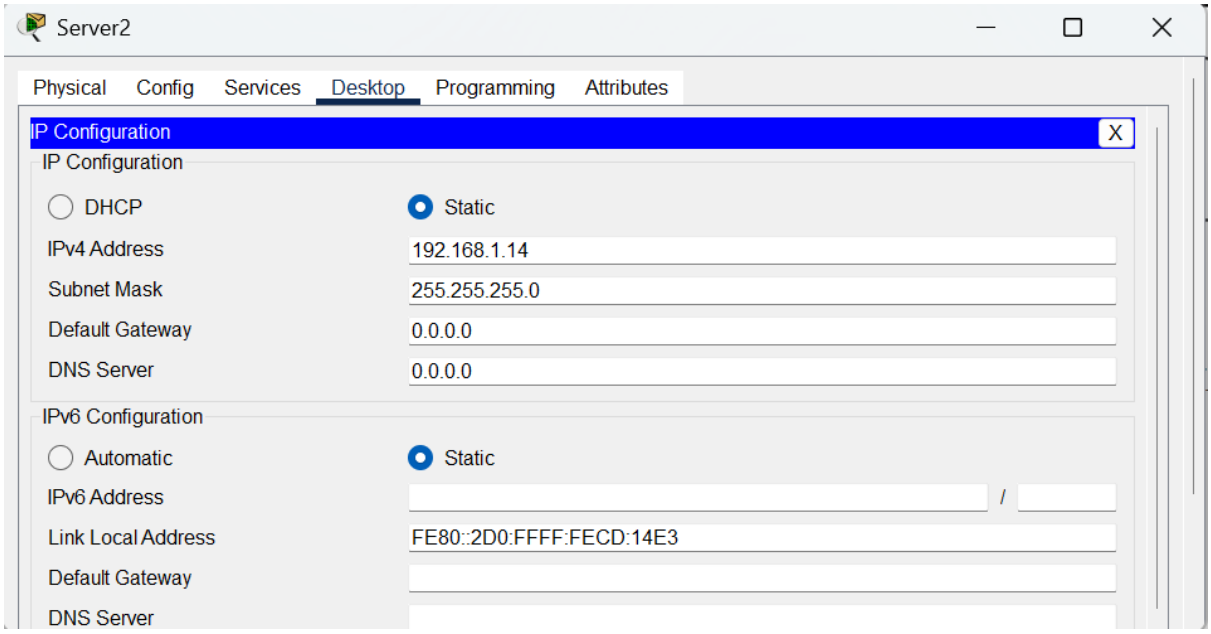

Switch2
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW3
SW3(config)#interface vlan 1
SW3(config-if)#ip address 192.168.1.3 255.255.255.0
SW3(config-if)#no shutdown
SW3(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
SW3(config-if)#exit
SW3(config)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console

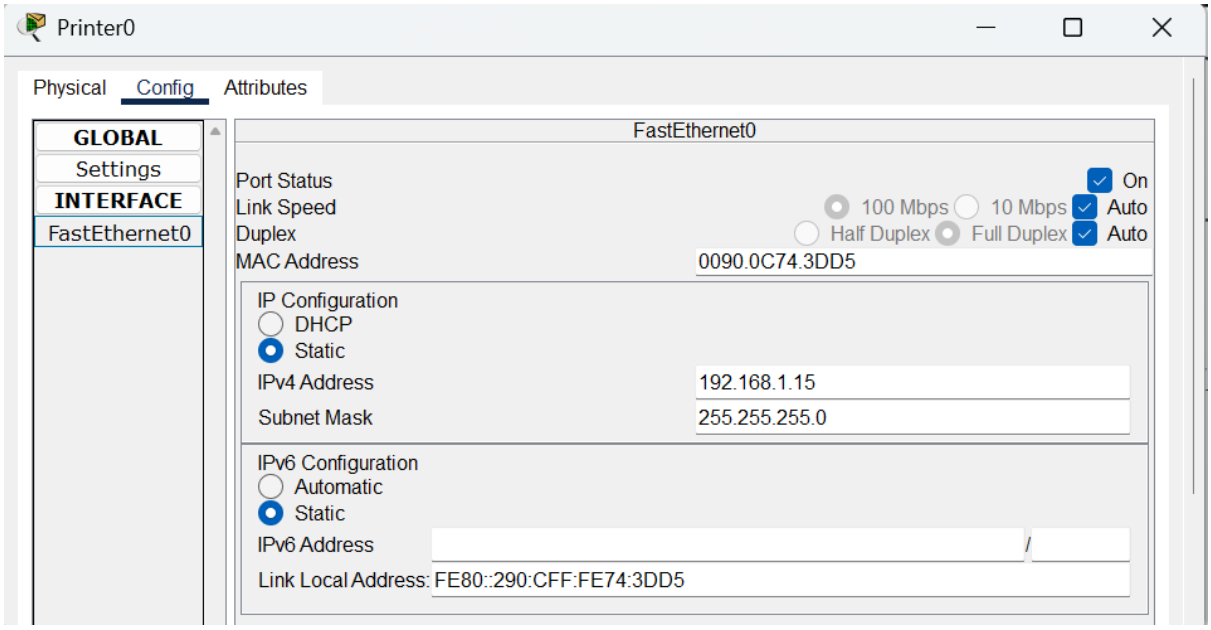



**Configuration des dispositifs**

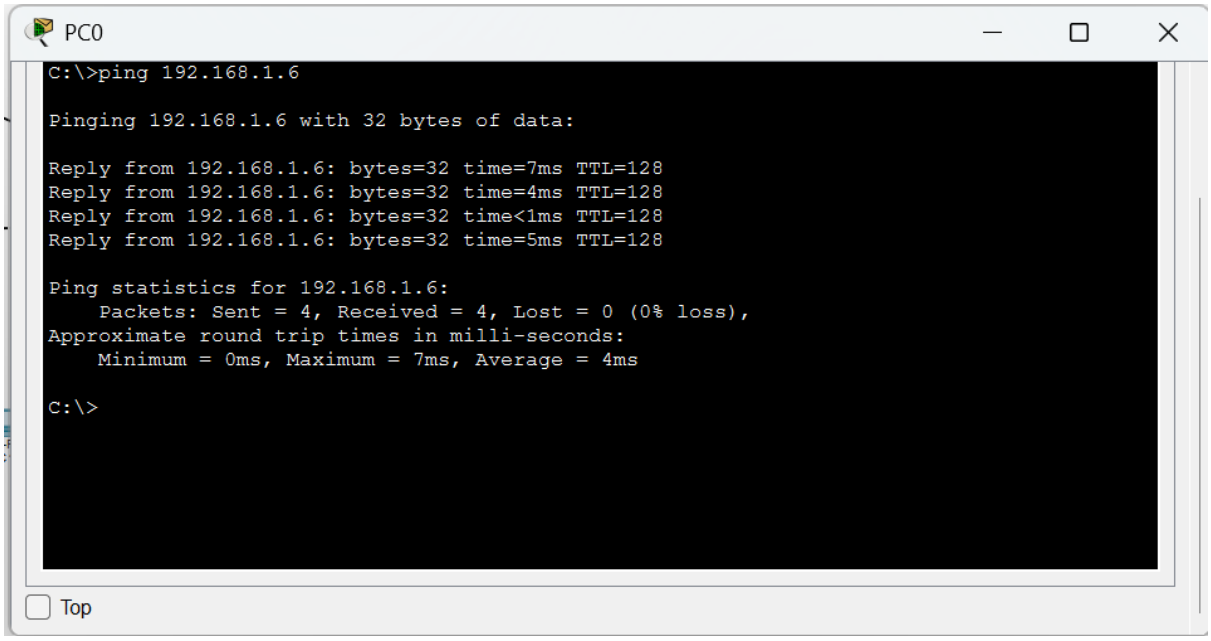


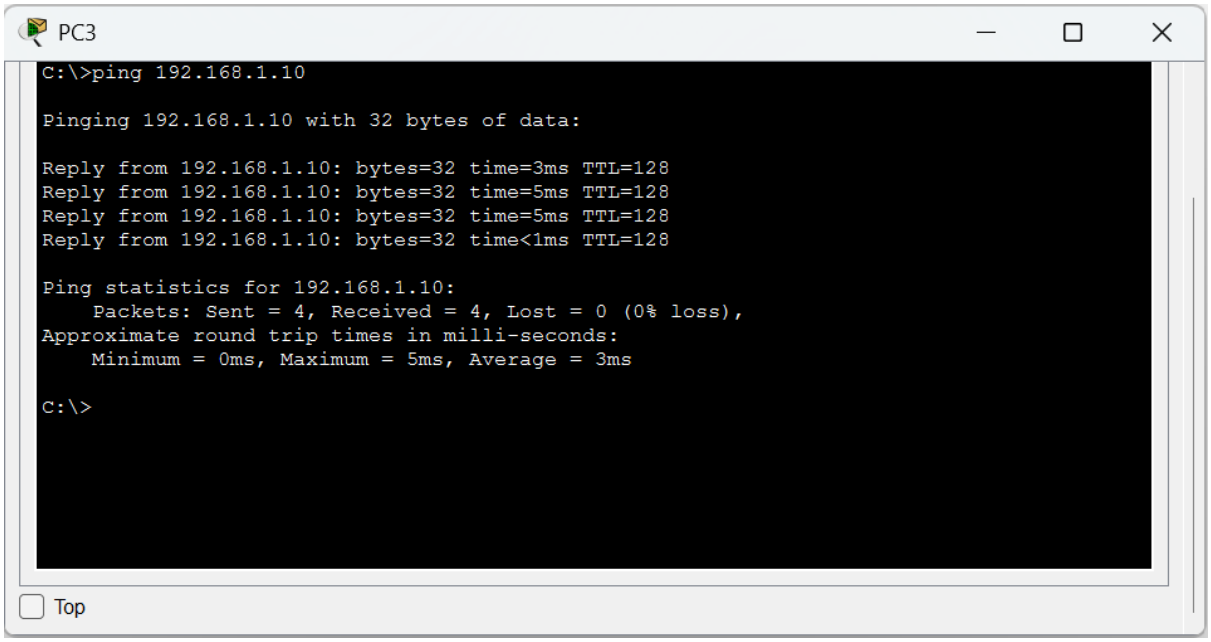

PC3
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::206:2AFF:FE2A:1EA5
Default Gateway
DNS Server
802.1X
Top

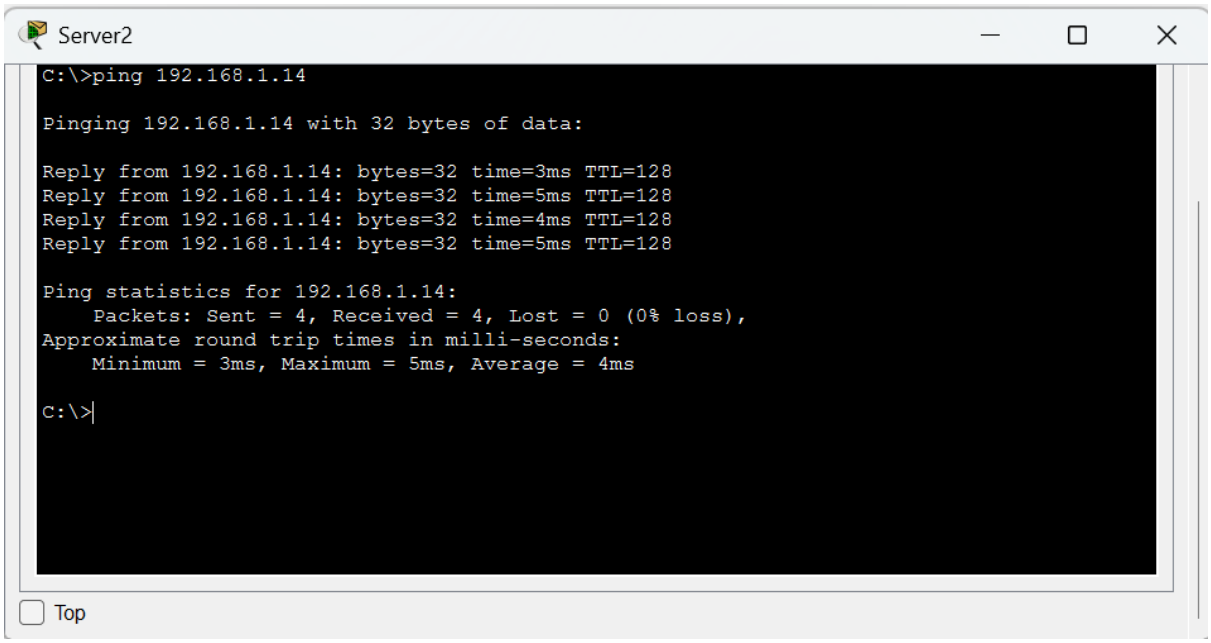

Server2
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.14
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:FFFF:FECD:14E3
Default Gateway
DNS Server



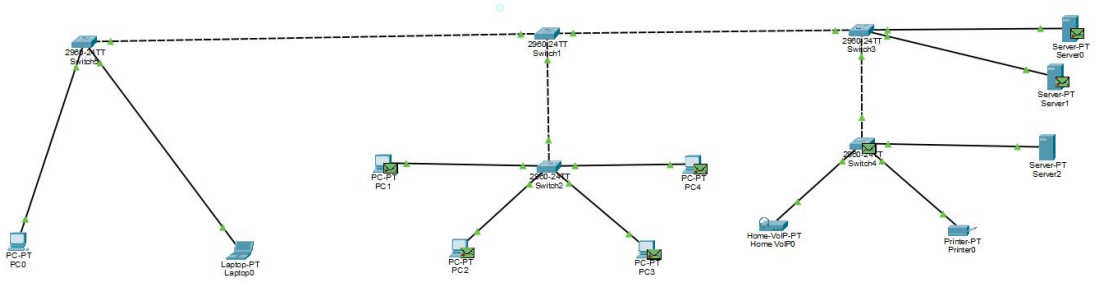

**Tester la connectivité entre les hôtes**

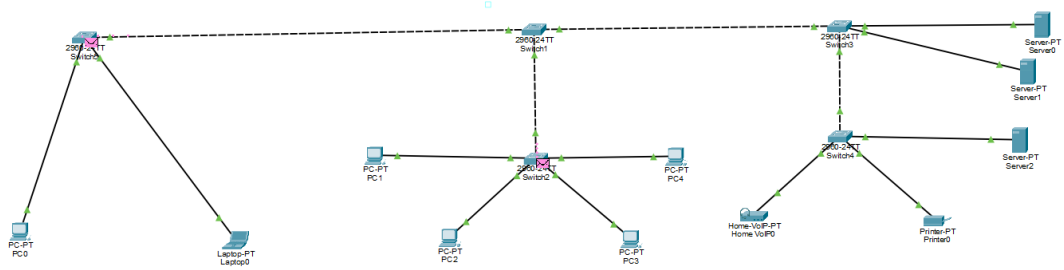




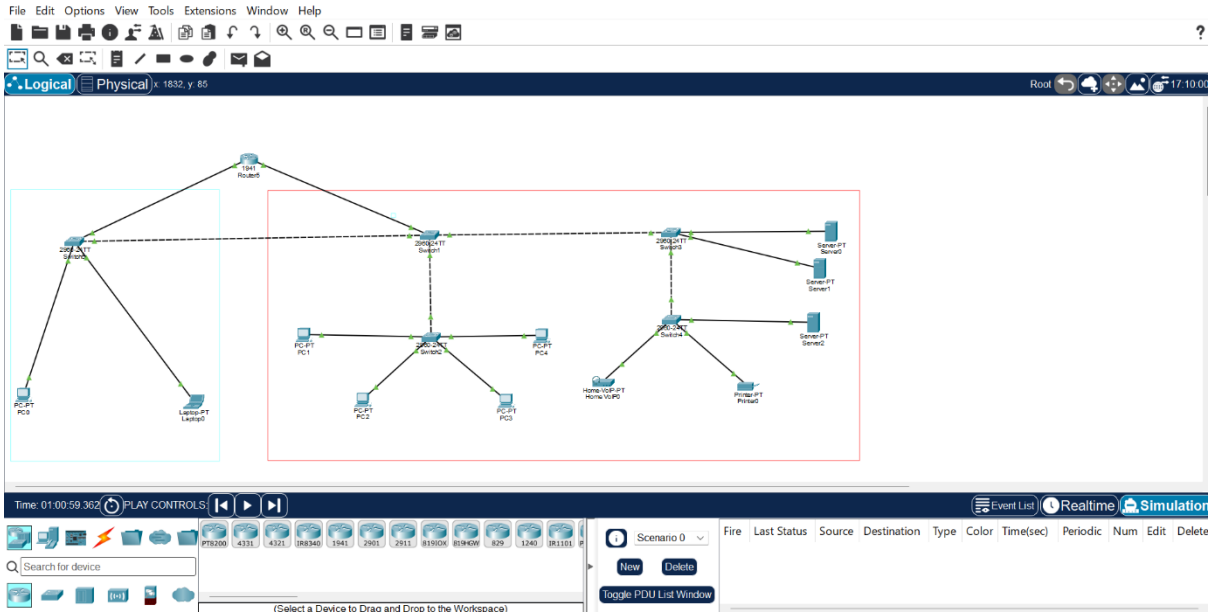




**Mode temps réel (Realtime Mode)**

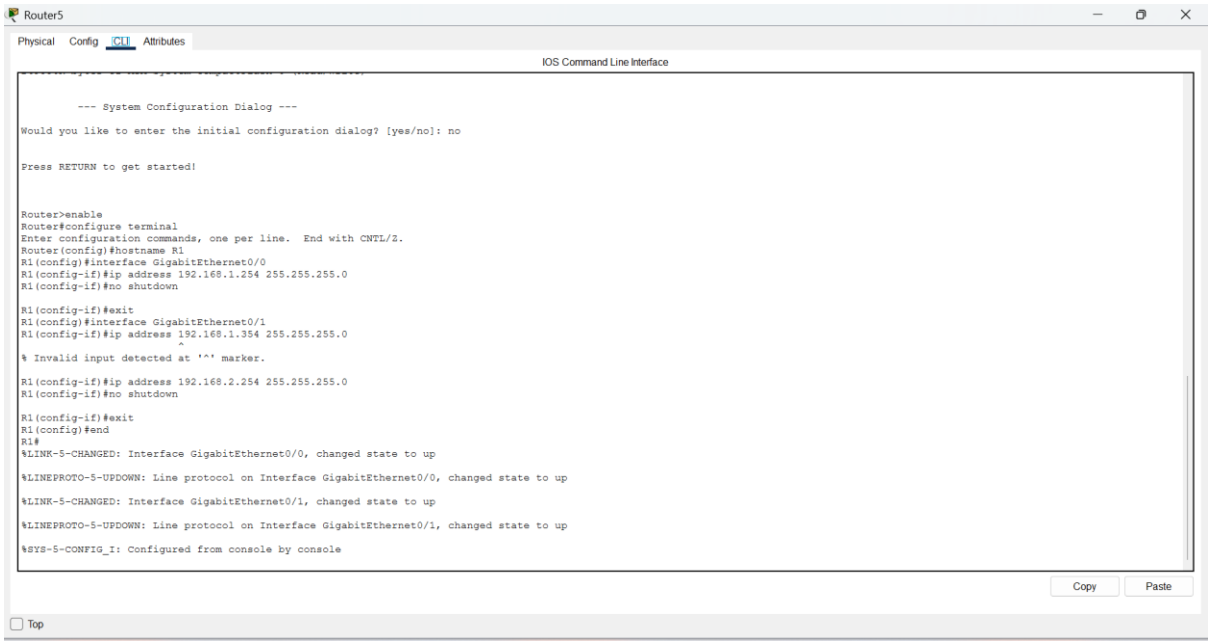


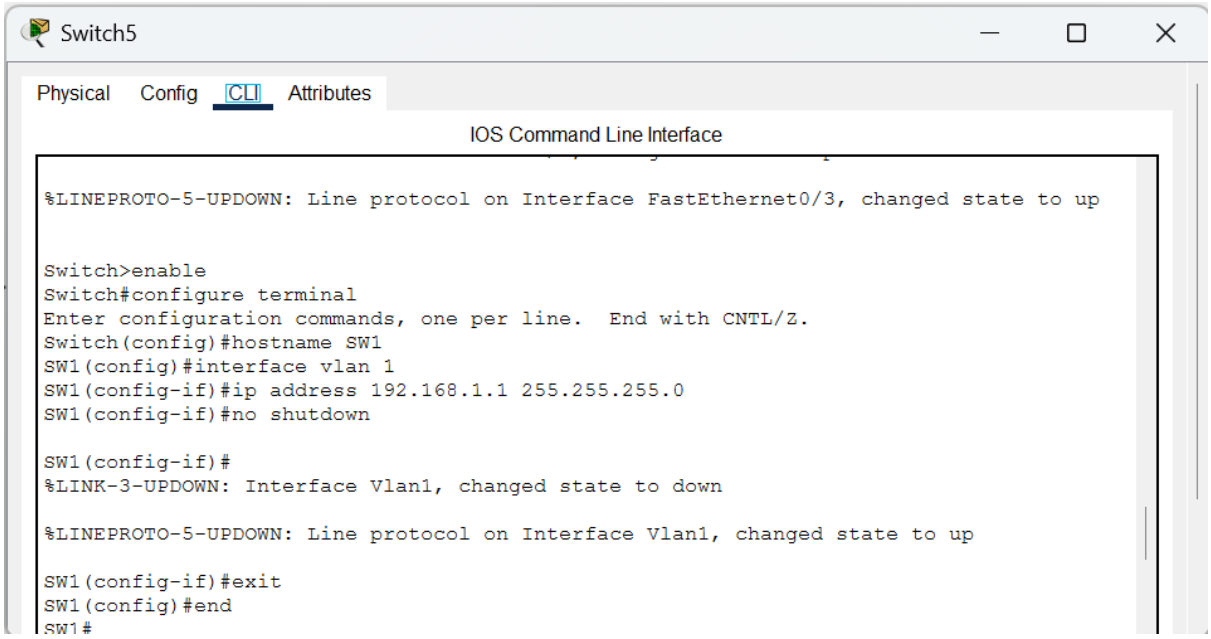

2. Reproduisez cette topologie en configurant le routeur et les switchs, puis en attribuant les adresses IP aux dispositifs. Utilisez soit IPv4, soit IPv6, et testez la connectivité des deux VLAN à l'aide de la commande ping et du mode de simulation.

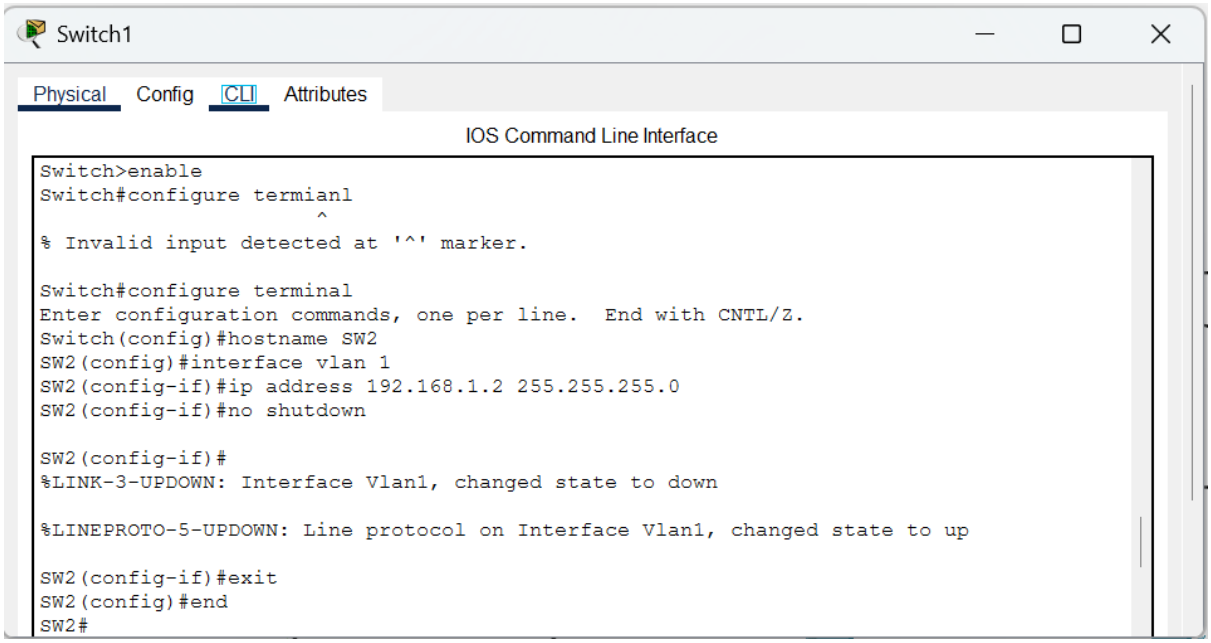


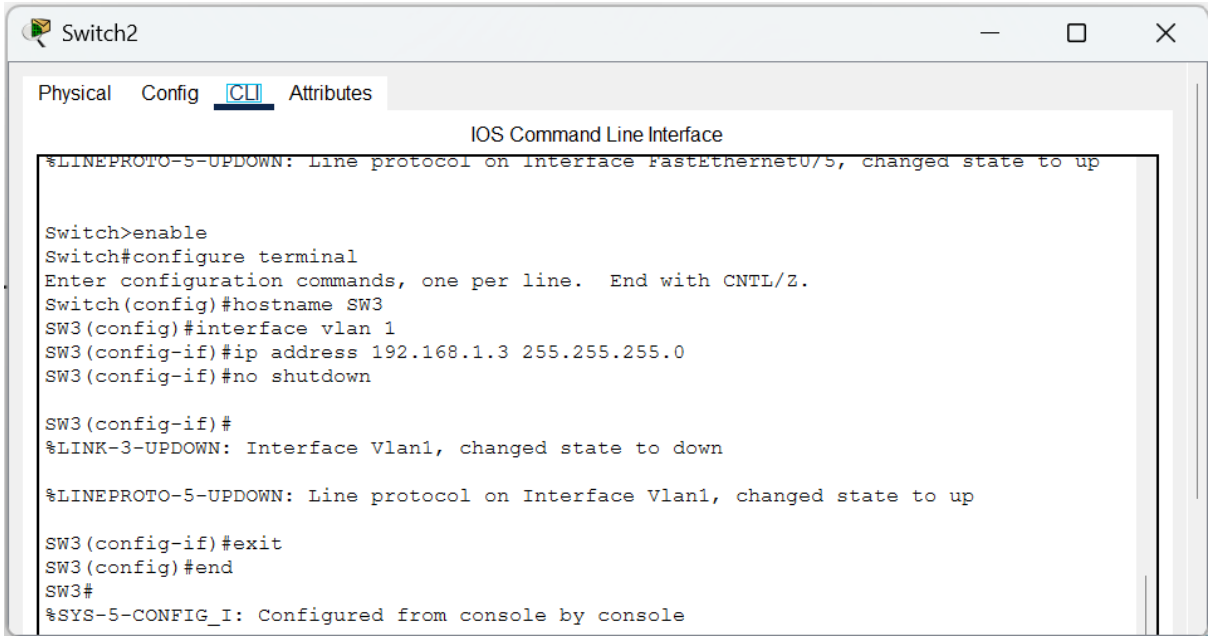

**Configuration du routeur**

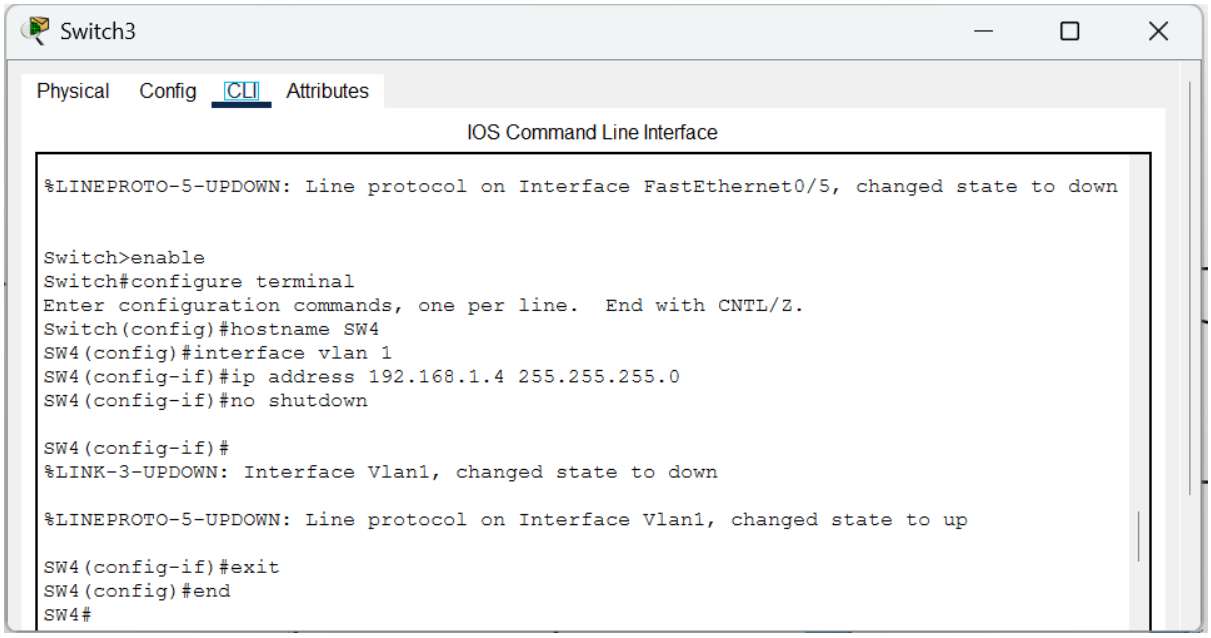



**Configuration des switchs IPv4 :**

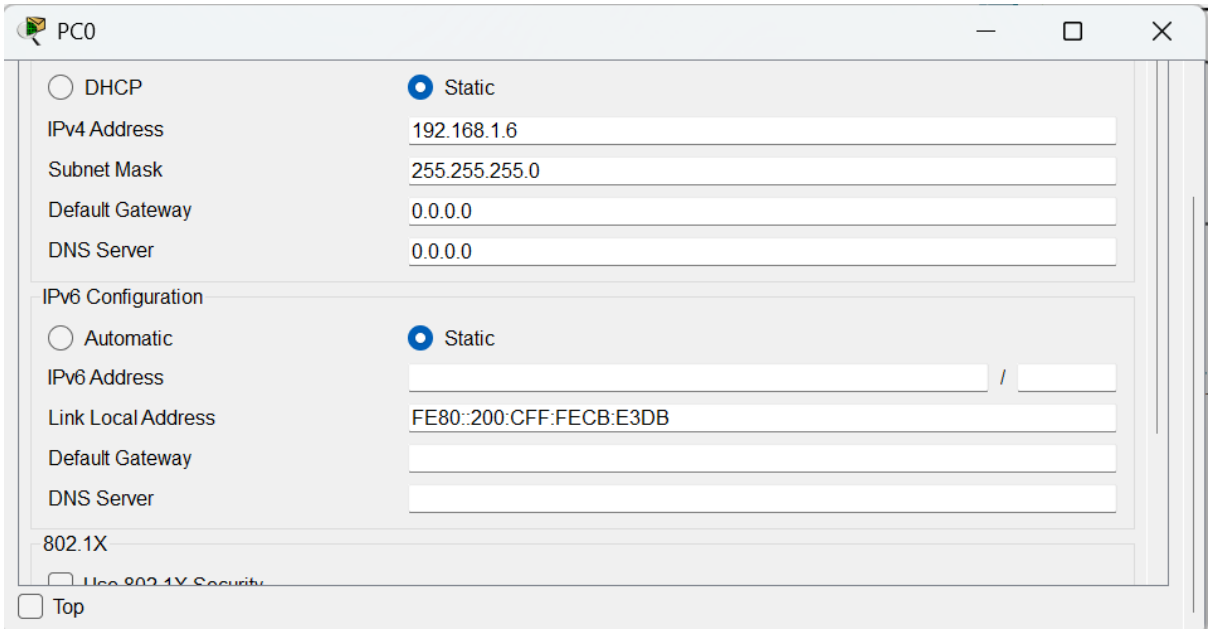

Switch5
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#interface vlan 1
SW1(config-if)#ip address 192.168.1.1 255.255.255.0
SW1(config-if)#no shutdown
SW1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
SW1(config-if)#exit
SW1(config)#end
SW1#

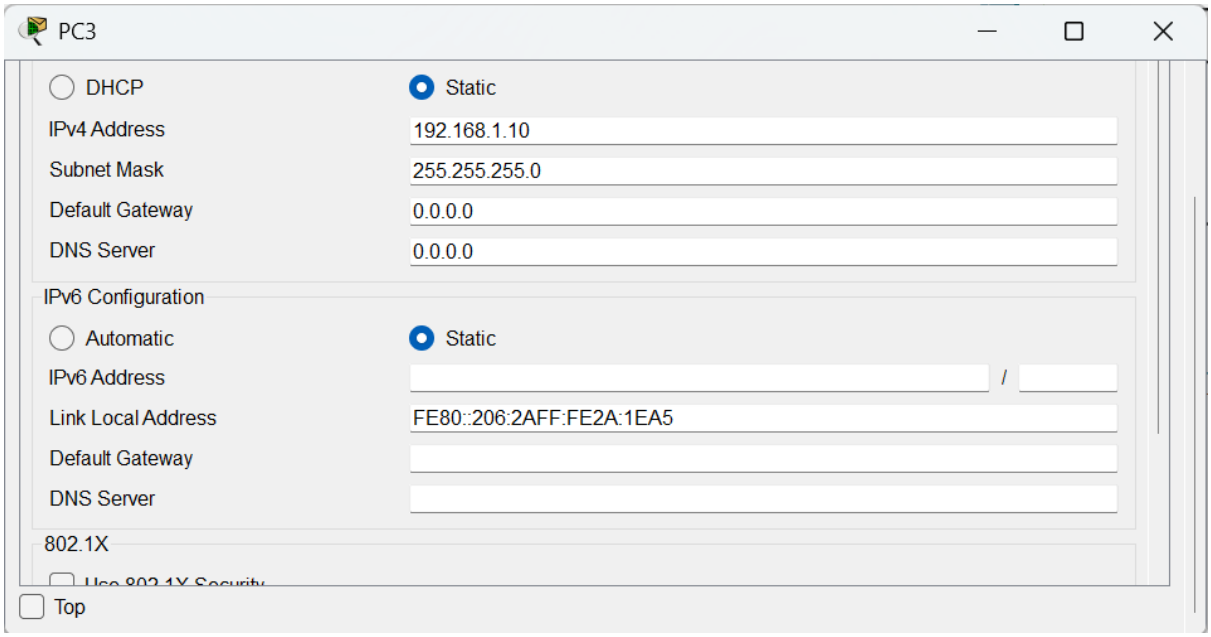

Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>enable
Switch#configure termianl
^
% Invalid input detected at '^' marker.
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW2
SW2(config)#interface vlan 1
SW2(config-if)#ip address 192.168.1.2 255.255.255.0
SW2(config-if)#no shutdown
SW2(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
SW2(config-if)#exit
SW2(config)#end
SW2#

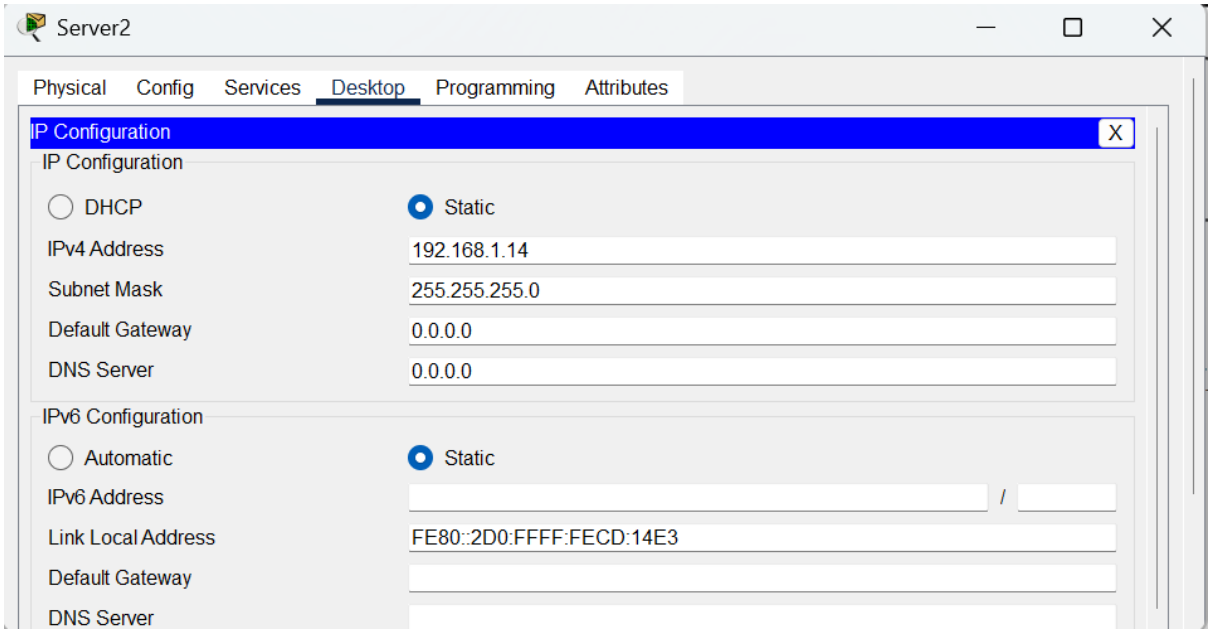


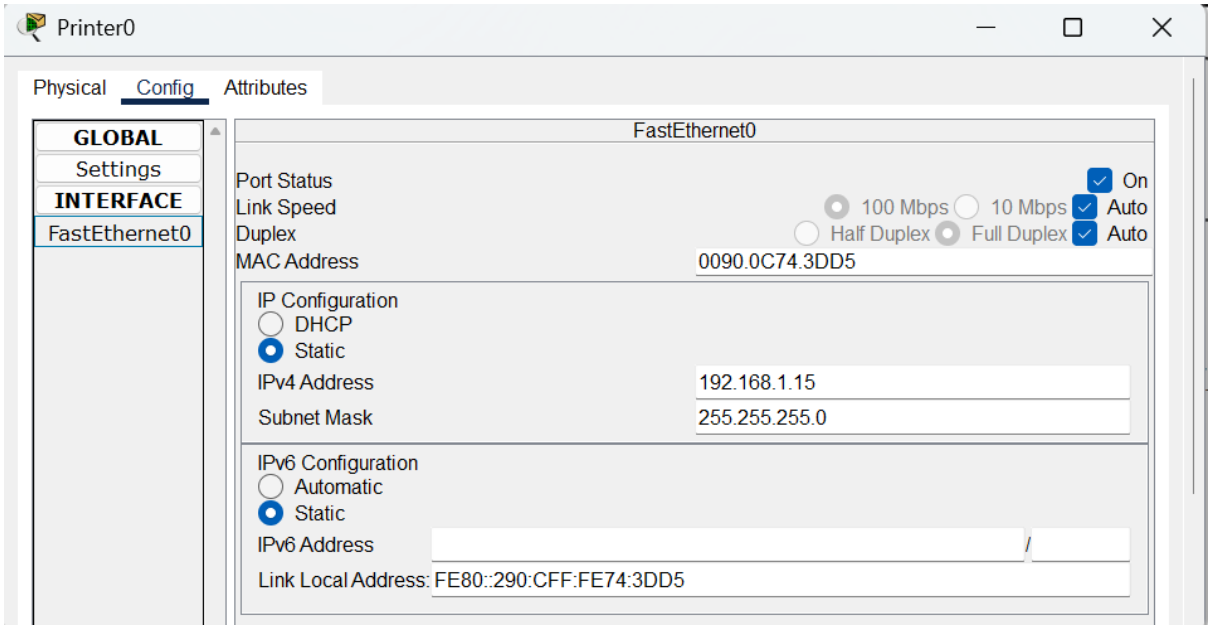




**Configuration des dispositifs :**

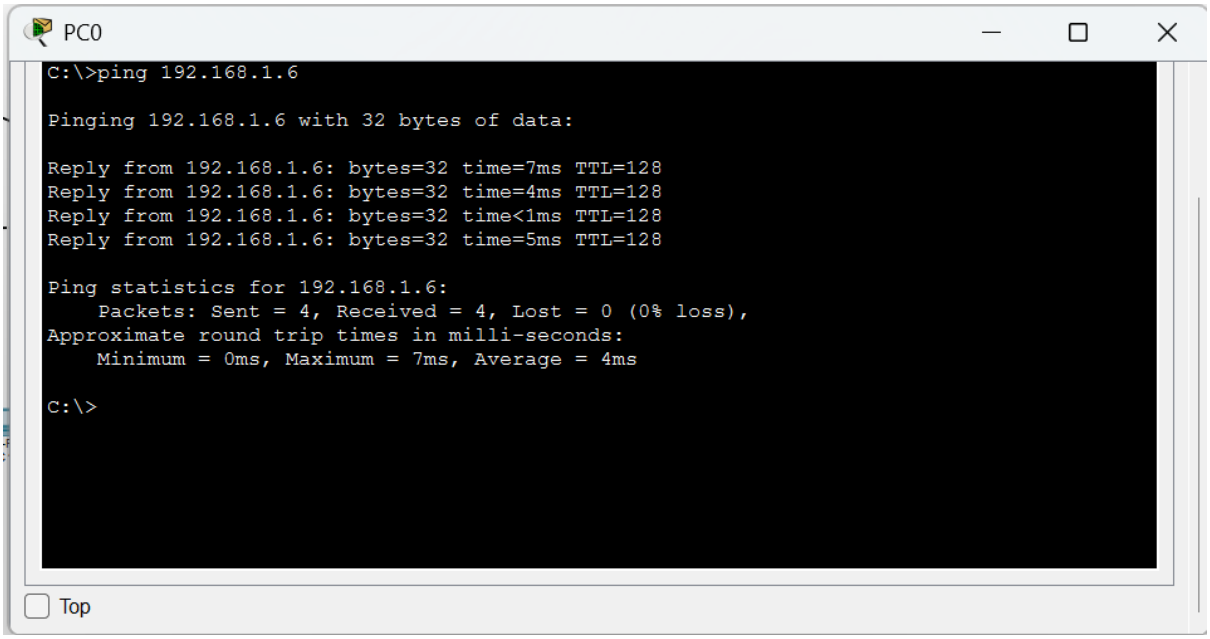

PC0
DHCP
Static
IPv4 Address
192.168.1.6
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::200:CFF:FECB:E3DB
Default Gateway
DNS Server
802.1X
Top

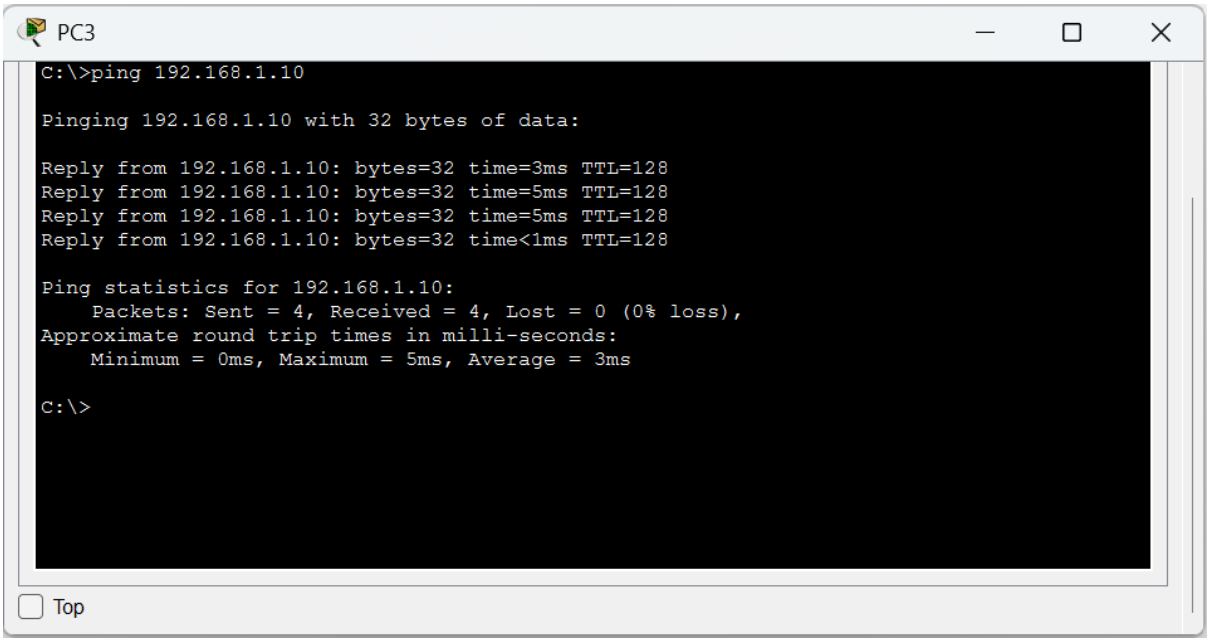

PC3
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::206:2AFF:FE2A:1EA5
Default Gateway
DNS Server
802.1X
Top

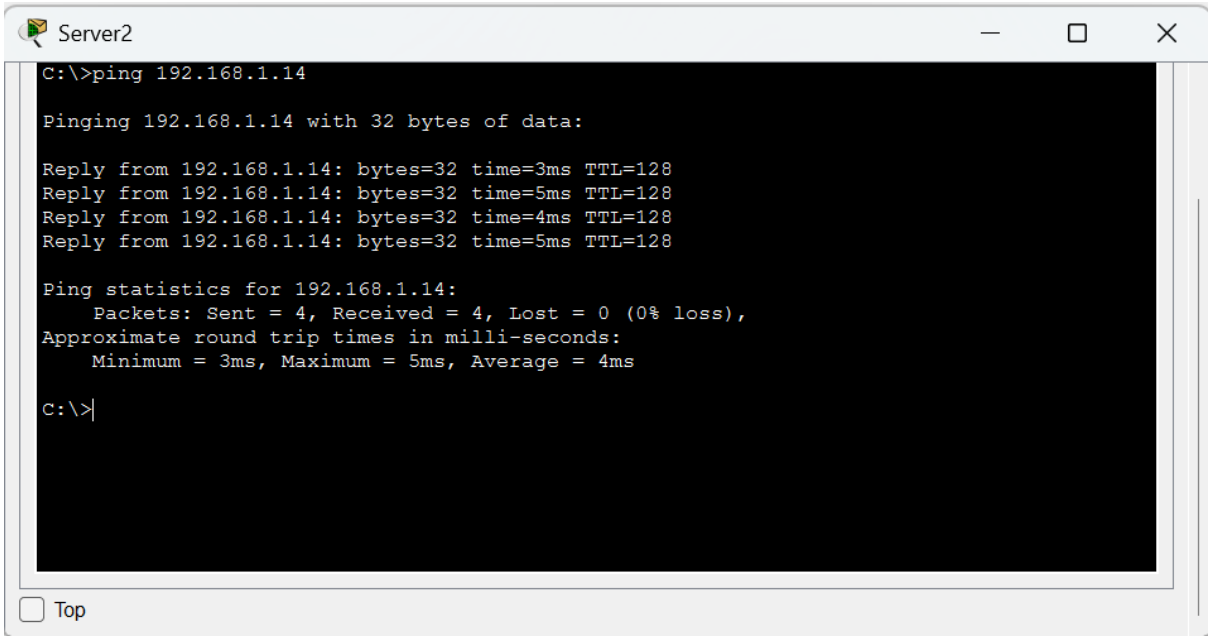


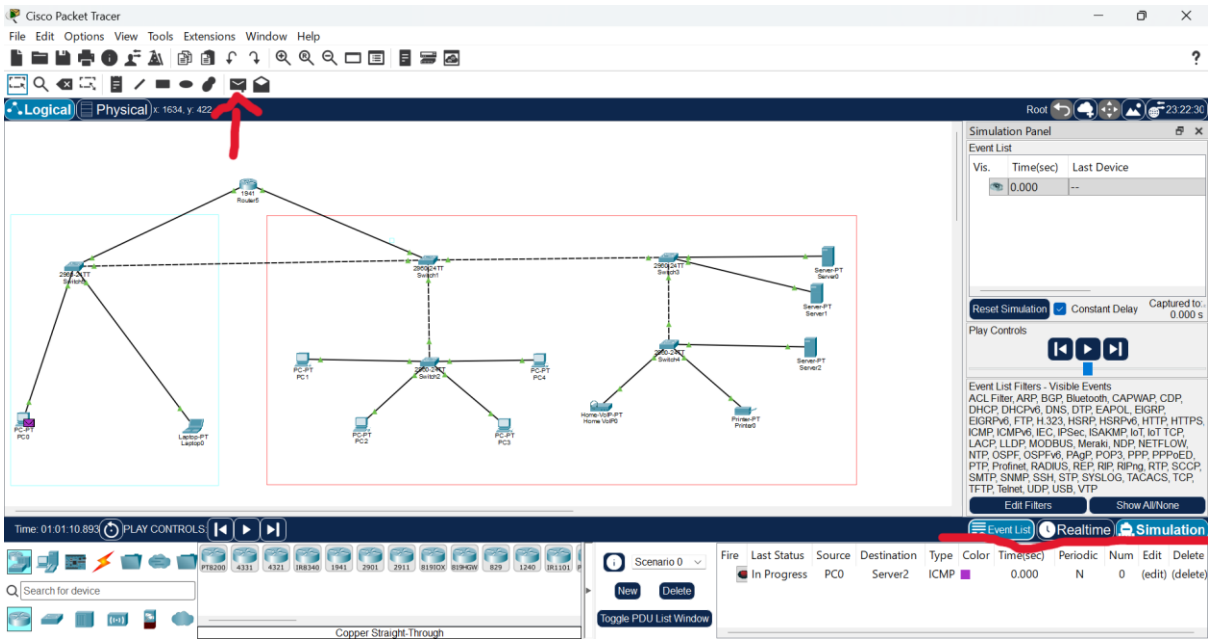

**Tester la connectivité entre les hôtes :**

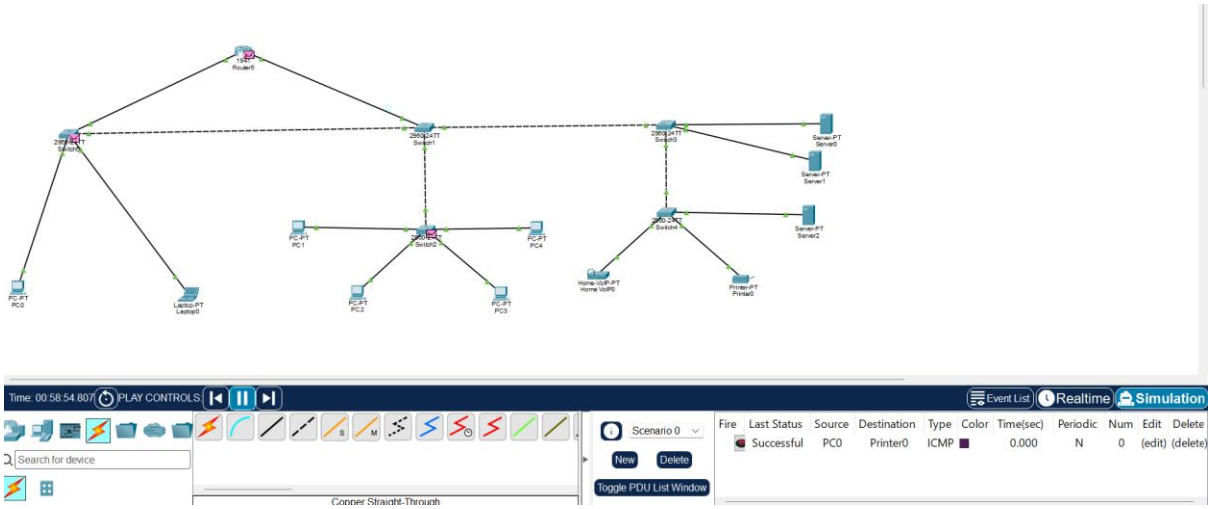

PC0
C:\>ping 192.168.1.6
Pinging 192.168.1.6 with 32 bytes of data:
Reply from 192.168.1.6: bytes=32 time=7ms TTL=128
Reply from 192.168.1.6: bytes=32 time=4ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time=5ms TTL=128
Ping statistics for 192.168.1.6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 7ms, Average = 4ms
C:\>
Top


PC3
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time=3ms TTL=128
Reply from 192.168.1.10: bytes=32 time=5ms TTL=128
Reply from 192.168.1.10: bytes=32 time=5ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 5ms, Average = 3ms
C:\>
Top




**Mode temps réel (Realtime Mode)**






**Conclusion**

Ce TD m'a permis de comprendre comment configurer un switch (nom, interface VLAN 1, adresse IP, passerelle par défaut) et un routeur (nom, interfaces, adresses IP, activation avec no shutdown). J'ai compris qu'un routeur relie des réseaux différents, que chaque interface doit avoir une adresse dans son propre réseau, et que les PC doivent avoir comme passerelle l'adresse de l'interface du routeur correspondant à leur réseau. J'ai aussi appris à vérifier la connectivité avec ping et à suivre le trajet des paquets en mode Simulation. La tâche a été réussie : les configurations ont été appliquées et les tests de connectivité entre les hôtes ont abouti.